Project Context

Project Title: The Paradigm Shift: From Model-Centric Scaling to Data-Centric Pipelines in Multimodal Video Contextualization

1. The Problem Statement (The Contextual Bottleneck)

The current trajectory of multimodal Artificial Intelligence relies heavily on the "Bitter Lesson" of scaling: adding more parameters to solve more complex problems. However, video contextualization introduces a severe mathematical and hardware wall. Because standard Transformer attention mechanisms scale quadratically, processing raw, high-resolution video frames causes an exponential explosion of visual tokens. This exhausts GPU VRAM, spikes inference latency, and renders real-time edge deployment mathematically impossible for massive Large Language Models (LLMs). The industry's reliance on model-centric scaling to brute-force this problem is computationally unsustainable.

2. The Core Hypothesis

This project hypothesizes that the true bottleneck in long-form video contextualization is not parametric reasoning capacity, but rather inefficient, noisy data-ingestion pipelines. The study posits that when visual perception is decoupled from textual reasoning—using data-centric, parallelized frame sampling and spatial-temporal compression—a Small Language Model (SLM, ~7B parameters) will demonstrate competitive reasoning capabilities against a massive LLM (~72B parameters) at a fraction of the computational cost.

3. The Methodology (Strict Variable Isolation)

To empirically prove this, the project will strictly isolate architectural bias (the NLP reasoning backbone) from data bias (the computer vision input).

- **The Control:** The project will compare open-weight models from the exact same family (e.g., Qwen2.5-VL-7B vs. Qwen2.5-VL-72B) that share an identical vision encoder.
- **The Pipeline:** Neither model will use its default, black-box preprocessing script. Both will be forced through an identical, custom-engineered frame-sampling pipeline built using optimized computer vision libraries (OpenCV/FFmpeg) to ensure the exact same array of pixels reaches the language backbone.
- **The Orchestration:** Inference loops will be optimized to maximize parallel processing, ensuring hardware latency does not skew the model evaluation metrics.

4. The Evaluation Matrix

Models will be benchmarked on a holistic matrix that measures both intelligence and systems-level efficiency:

- **Task Accuracy:** Measured against standardized datasets like Video-MME.
- **Computational Cost:** Time to First Token (TTFT), Tokens Per Second (TPS), and Peak

VRAM utilization.

5. The Ultimate Goal

To engineer a reproducible, highly documented multimodal evaluation pipeline that proves data engineering and input curation are mathematically superior to sheer parameter scaling for video AI.

Literature Extraction Framework

This combined framework synthesizes the goals of all three of your research segments into one master template. Apply this to every paper you read to systematically extract exactly what you need for your code, your workflow, and your thesis defense.

I. Meta-Data & Core Classification

- **Paper Title & Year:** * **Primary Focus:** Is this paper strictly *Data-Centric* (optimizing inputs, SLMs, pipelines) or *Model-Centric* (scaling parameters, MoE, FLOPs)?
- **The Paradigm Shift:** How does this specific paper challenge or support the traditional "bigger is better" scaling laws?

II. The Theoretical Foundation (The "Why")

- **Mathematical Limits:** Does the paper define specific computational limits? (e.g., the optimal token-to-parameter ratio, or the point where attention mechanisms collapse under video length).
- **The Bottleneck Addressed:** What specific failure point does this address? (e.g., temporal blindness, catastrophic forgetting, modality interference, or VRAM exhaustion).
- **Definitions:** Are there any specific academic terms (e.g., *Inference Compute-Optimal*, *Router Rigidity*) that need to be added to the thesis glossary?

III. Methodological Blueprint (The "How")

- **Data Pipeline Mechanics:** What exact techniques are used to format the video before model ingestion? (e.g., uniform frame sampling rates, extracting bounding boxes, spatial-temporal grid stitching, or generating synthetic captions).
- **Prompt Engineering & Orchestration:** What specific system prompts or iterative agentic loops are used to force the model into chronological, step-by-step temporal reasoning?
- **Architectural Alterations:** If model-centric, what structural changes were made? (e.g., introducing Mamba blocks, applying 4-bit AWQ quantization, or creating Mixture-of-Expert routing layers).

IV. Project Integration & Actionability (The Deliverables)

- **Control Variables:** What specific parameters from this paper must be replicated in Phase

3 (Data Engineering) to ensure the SLM vs. LLM comparison remains fair and scientifically valid?

- **Hardware Requirements:** What VRAM limits, parallel computing libraries (like vLLM), or Python dependencies does this paper prove are necessary for Phase 2 (Environment Setup)?
- **Handling Failure Modes:** Does this paper outline how models typically fail (e.g., hallucinating code, crashing on corrupted frames) so that specific try-except error handling can be coded into the automated inference loop?

V. Synthesis & Defense Preparation

- **The Distinguishing Factor:** How does this study differentiate itself from generic image-based benchmarks or older, flawed video studies?
- **Thesis Contribution:** Where does this extraction belong in the final document? Does it justify the frame extraction script, defend the quantization choice, or provide a theoretical counter-argument for the "Limitations" chapter?